Language Models are Unsupervised Multitask Learners (Radford et al. 2019)

Autoregressive language models trained purely on diverse web text demonstrate zero-shot task transfer across reading comprehension, translation, summarization, and question answering without explicit supervision.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Training an autoregressive Transformer (GPT-2, up to 1.5B parameters) on a large, diverse dataset of web text without explicit task supervision allows the model to perform multiple NLP tasks in a zero-shot setting simply by predicting the next token.

EVIDENCE AND CAUSAL ASSESSMENT

Evaluations across four model sizes (117M, 345M, 762M, 1.5B) isolate parameter capacity as the causal driver of zero-shot task performance, with capabilities improving log-linearly with scale.

EFFECT SIZE AND PERSISTENCE

GPT-2 achieved state-of-the-art results on 7 out of 8 tested language modeling benchmarks in a zero-shot setting, demonstrating competitive performance on summarization, translation, and question answering without task-specific training data.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%). Web-scraped training datasets contain substantial demographic and factual biases, and zero-shot task performance lags behind task-specific supervised baselines on complex reasoning benchmarks.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This does not mean that the model was secretly fine-tuned; rather, it proves that the objective of next-token prediction over broad distributions contains diverse downstream tasks as sub-problems.

WHAT ACTION IS REASONABLY JUSTIFIED

Leverage large-scale generative pre-training as the universal initialization substrate for language tasks before considering specialized supervised fine-tuning.

VERIFIED SOURCES & DOI IDENTIFIERS

- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners. OpenAI Technical Report.